

Dilations and Similar Figures

Dilating coordinates, finding scale factors, checking slopes, and scaling lengths

Directions

- Every dilation on this sheet is centred at the origin. To dilate a point by scale factor k , multiply **both** coordinates by k : $(x, y) \mapsto (kx, ky)$.
- Lengths taken from the coordinate grid are in grid units. Real-world lengths carry the unit named in the question — write it on the answer line.
- Show the calculation you used, not just the result.

1. The point $A(2, 3)$ is dilated about the origin with scale factor $k = 3$.

$A'(\text{_____, _____})$

2. $P(1, 2)$ and $Q(4, 6)$ are dilated about the origin with scale factor $k = 2$, giving $P'(2, 4)$ and $Q'(8, 12)$.

Length of $PQ = \text{_____}$ Length of $P'Q' = \text{_____}$ (grid units)

3. A rectangular photo print measures 6 cm along its short side and 9 cm along its long side. An enlargement of the print is similar to it and measures 15 cm along its short side.

(a) Scale factor $k =$ _____

(b) How long is the long side of the enlargement?

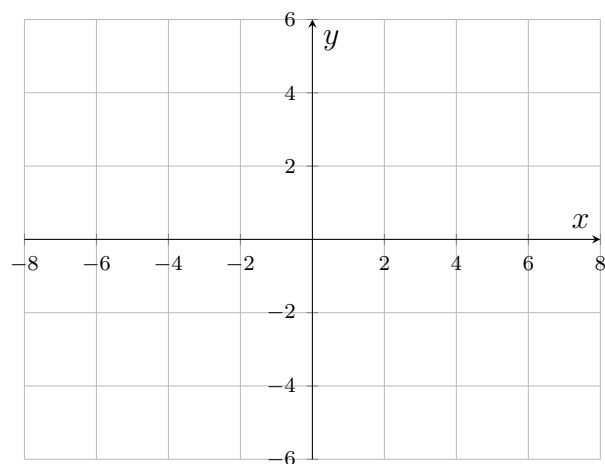
Answer: _____ cm

4. $A(1, 2)$ and $B(4, 8)$ are dilated about the origin with scale factor $k = \frac{1}{2}$, giving $A'(0.5, 1)$ and $B'(2, 4)$.

Slope of $AB =$ _____ Slope of $A'B' =$ _____

Explain what those two slopes tell you about the segments AB and $A'B'$.

- 5.** Triangle ABC has vertices $A(-3, -1)$, $B(1, -1)$ and $C(1, 2)$. Dilate it about the origin with scale factor $k = 2$ and draw the image triangle $A'B'C'$ on the grid, labelling each image vertex.



Length of the image side $A'B' =$ _____ (grid units)

- 6.** A photograph is 8 cm wide and 12 cm tall. It is enlarged so that the enlargement is similar to the original and is 20 cm wide. How tall is the enlargement?

Answer: _____ cm

7. $D(2, 1)$ and $E(4, 5)$ are dilated about the origin, giving $D'(6, 3)$ and $E'(12, 15)$.

Slope of DE = _____ Slope of $D'E'$ = _____ Scale factor k = _____

8. Triangle RST has $R(0, 0)$, $S(4, 0)$ and $T(0, 3)$. A student claims that the triangle with vertices $(0, 0)$, $(8, 0)$ and $(0, 9)$ is the image of RST under a dilation centred at the origin.

Factor along RS = _____ Factor along RT = _____

Is the student right? Explain, using both factors.

9. A scale model of a bridge is similar to the real bridge, and every real length is 40 times the matching model length. The model's main span measures 3.2 m and its tower measures 0.45 m.

(a) How long is the real main span?

Answer: _____ m

(b) How tall is the real tower?

Answer: _____ m

10. Triangle JKL has $J(0, 0)$, $K(6, 0)$ and $L(0, 8)$. It is dilated about the origin with scale factor $k = 1.5$.

(a) Length of $KL =$ _____ (b) Perimeter of $JKL =$ _____ (c) Perimeter of $J'K'L' =$ _____

(d) Explain how part (c) can be found from part (b) without working out a single image coordinate.
